

ECO101H1F L101 & L201: Principles of Microeconomics
Midterm 1 (Version A), Fall 2024

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Multiple Choice Questions

PUT MULTIPLE CHOICE ANSWERS IN THE BUBBLE SHEET (LAST PAGE).

No work here will be considered.

- 1) [2 points] Your values expressed in dollars from three entertainment options tonight are \$80 for a concert, \$70 to watch a tennis match, and \$60 for dinner. All of them are priced at \$40. You can go to only one. Based on this information, what is the opportunity cost of going to dinner?
- A. \$120
 - B. \$80
 - C. \$70
 - D. \$40
 - E. None of the above.

Solution: B. If you go to the dinner, your next best option is going to the concert. Your Opportunity Cost is the \$40 you spend plus the net benefit you give up from the concert: $\$40 + (\$80 - \$40) = \80 .

- 2) [2 points] Alfonso has downward-sloping demand for manga comics. At a price of \$15 a comic he buys 14 comics a month. If the price of a comic drops to \$10, his marginal willingness to pay (MWTP) for the 14th comic...
- A. decreases by \$5
 - B. decreases by less than \$5
 - C. increases by \$5
 - D. increases by less than \$5
 - E. stays the same

Solution: E. His quantity choice will increase, because more units will pass the MWTP-versus-price threshold, but his MWTP for each individual unit is a feature of his preferences and stays exactly the same.

- 3) [2 points] Both Ianville and Jonland benefit when Ianville sends bananas in exchange for apples from Jonland. Based only on this information, ____ has the absolute advantage in banana production and ____ has the comparative advantage in apple production.
- A. Ianville; Jonland
 - B. Jonland; Ianville
 - C. either could have; Jonland
 - D. either could have; Ianville
 - E. None of the above.

Solution: C. Since both benefit from trade, we know that Jonland has the lowest Opportunity Cost in apple production and hence has the comparative advantage there. However, we aren't told how productive either is at either fruit, so we cannot say anything about absolute advantage.

- 4) [2 points] Assume linear PPFs. Workers in York work 6 days per week, producing either wine or juice with their time. Recently York passed a law reducing the number of days each worker works from 6 to 5 days a week, all else stays the same. As a result, in York, the Opportunity Cost of an extra juice will ____ and the maximum amount of juice it can produce will ____.

- A. increase; decrease.
- B. decrease; stay the same.
- C. could increase or decrease; decrease.
- D. stay the same; increase.
- E. stay the same; decrease.

Solution: E. A reduction in resources alone moves the PPF inwards, so it decreases the maximum amount that can be produced. But it doesn't change relative productivities — the slope of the PPF is unchanged — so it does not change the Opportunity Cost.

- 5) [2 points] Each buyer buys only 1 unit if they buy the good. Their willingness to pay (WTP) for a unit is given in the table below. If the market price is \$17 per unit, how many units will be demanded?

Buyer	A	B	C	D	E
WTP	\$6	\$27	\$13	\$21	\$12

- A. 1 unit
- B. 2 units
- C. 3 units
- D. 4 units
- E. 5 units

Solution: B. Only B (\$27) and D (\$21) have a WTP of \$17 or higher, which means only 2 units will be demanded.

- 6) [2 points] When the market price rises from \$8 to \$12, market quantity demanded decreases from 900 to 300 units. Based on this information, the (absolute value of) elasticity of demand is ____.

- A. 0.4
- B. 0.75
- C. 1.33
- D. 2.5
- E. None of the above.

Solution: D. Quantity decreases by 600 units when price increases by \$4. Using midpoints as the base,

$$|\varepsilon_D| = \left| \frac{-600/600}{4/10} \right| = 2.5$$

- 7) [2 points] Assume a competitive market for vibranium. The (absolute value of) price elasticity of demand for vibranium is 1.5 and the income elasticity is equal to 2. If there is an increase in income then:

Statement 1: The price of vibranium will increase.

Statement 2: Total revenue will increase.

Based on the information in the question, Statement 1 is ____ and Statement 2 is ____.

- A. TRUE; TRUE

- B. TRUE; FALSE
- C. FALSE; TRUE
- D. FALSE; FALSE
- E. We need more information to answer the question.

Solution: A. Since the income elasticity is positive, an increase in incomes shifts demand out, moving us up along the supply curve to a higher price and a higher quantity. A higher price means Statement 1 is true. As both P and Q increase, revenues increase as well, so Statement 2 is true too. The own-price elasticity of 1.5 is a red herring — we are moving along supply, not along demand.

- 8) [2 points] Assume no externalities. For the last available ECO101 Stanley Quencher, Polina was willing to pay \$70 but actually paid \$40. Avner is willing to pay \$90 and Niobe is willing to pay \$120. If Polina sells the Quencher to Avner for \$80, total surplus...
- A. decreases by \$30
 - B. stays the same
 - C. increases by \$10
 - D. increases by \$20
 - E. None of the above.

Solution: D. When the mug is reallocated away from Polina, society loses the \$70 of benefit she would have received. When Avner gets the mug, society gains the \$90 of benefit he gets from having it. The net is an increase of \$20. The \$80 price and the \$40 Polina paid don't enter — one is a transfer, the other is sunk. (Niobe's \$120 would have been better still, but she isn't the one buying.)

- 9) [3 points] Xiaobo's total benefit from the number of weekly pickleball lessons is given in the table below. He is given a choice between paying \$30 per lesson or paying \$100 per week for all the lessons he wants. How many lessons does he take a week?

Number of lessons	1	2	3	4	5
Total Benefit	\$80	\$140	\$180	\$200	\$190

- A. 1 lesson
- B. 2 lessons
- C. 3 lessons
- D. 4 lessons
- E. 5 lessons

Solution: D. If he pays \$30 a lesson he chooses 3 lessons, comparing $MC = \$30$ against the Marginal Benefits in the table below, for a surplus of $\$180 - \$90 = \$90$. If he chooses \$100 per week, his $MC = \$0$ for an extra lesson and so he chooses 4 lessons, for a surplus of $\$200 - \$100 = \$100$. That is higher, so he takes the weekly deal and 4 lessons.

Number of lessons	1	2	3	4	5
Marginal Benefit	\$80	\$60	\$40	\$20	-\$10

- 10) [3 points] It takes Brutus 2 hours to walk to or from work. Walking is free. The price of taking the TTC is \$3.50 each way, and if he takes the TTC it takes 30 minutes either direction. Suppose construction delays increase the TTC commute to 1 hour each direction, all else stays the same. As a result, the Opportunity Cost of taking the TTC ____ and Brutus' willingness to pay for the TTC ____.
- A. decreases; decreases

- B. decreases; increases
- C. increases; decreases
- D. increases; stays the same
- E. None of the above.

Solution: C. His Opportunity Cost of taking the TTC is the fare plus the time it takes, plus the net value of the walk he gives up:

$$OC_{TTC} = \$time_{TTC} + \$3.50 + \$V_{walk} - \$2 \text{ hrs}$$

His Willingness to Pay is the largest fare at which he would still take the TTC, so setting $\$V_{TTC}$ equal to that Opportunity Cost and solving for the fare:

$$WTP_{TTC} = \$V_{TTC} - \$V_{walk} + \$2 \text{ hrs} - \$time_{TTC}$$

When the time taken by the TTC increases, $\$time_{TTC}$ rises. It enters the Opportunity Cost with a plus and the Willingness to Pay with a minus, so his Opportunity Cost increases and his Willingness to Pay decreases.

- 11) [3 points] Leia's Total Costs of making a quantity of bubble teas is given in the table below. If she can sell them at a price of \$10 each, how many bubble teas will she choose to produce?

Quantity	1	2	3	4
Total Cost	\$5	\$16	\$23	\$36

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

Solution: D. At the second unit $MB = P < MC$ (\$10 against \$11), but producing the third unit gives her an additional \$3 net, as $P > MC$ there (\$10 against \$7). She doesn't want to go on to the 4th unit, as $MB < MC$ (\$10 against \$13). Because Marginal Cost is not monotonic here, you have to check the totals rather than stop at the first unit where $MC > P$: net benefit is highest at 3 units.

Q	1	2	3	4
Total Cost	\$5	\$16	\$23	\$36
Marginal Cost	\$5	\$11	\$7	\$13
Total Benefit	\$10	\$20	\$30	\$40
Net benefit	\$5	\$4	\$7	\$4

- 12) [3 points] Assume a competitive market for paintings. To make a painting, Rabia needs time, canvas, and paint. The market price for a painting is \$50. If she doesn't paint, Rabia works at Metro as a cashier. Suppose (i) the price of canvas decreases, and (ii) Metro increases her hourly wage. As a result, if the market price of a painting remains \$50, the quantity of paintings sold by Rabia _____
- A. stays the same.
 - B. increases
 - C. either increases or stays the same.
 - D. decreases

E. could increase or decrease

Solution: E. Canvas is an input into her paintings, so a decrease in the price of canvas decreases her MC. Time is also an input, and when the Opportunity Cost of her time increases it increases her MC. We don't know the net effect, so we cannot determine how her quantity supplied changes.

- 13) [3 points] Octavio and Abby each have 10 hours per day they can devote towards cooking meals or washing dishes. Their productivity is given in the table below. They trade tasks. Which of the prices below for one meal will allow both to benefit from the trade?

	Cooking Meals	Washing Dishes
Octavio	6 meals per hour	3 dishes per hour
Abby	12 meals per hour	18 dishes per hour

- A. $\frac{1}{3} = 0.33$ washed dishes
 B. $\frac{2}{5} = 0.4$ washed dishes
 C. 1 washed dish
 D. $\frac{4}{2} = 2$ washed dishes
 E. 2.5 washed dishes

Solution: C. The Opportunity Cost of meals is given in the table below. Based on this, Octavio will sell meals at a minimum price of 0.5 washed dishes and Abby will buy meals at a maximum price of 1.5 washed dishes, so the price has to lie strictly between the two. Only 1 washed dish does.

	OC of Cooking Meals
Octavio	$\frac{3}{6} = \frac{1}{2}$ dishes
Abby	$\frac{18}{12} = 1.5$ dishes

- 14) [3 points] Assume perfectly competitive markets with the usual demand and supply curves. Beef and leather are produced from different parts of cattle. If the demand for beef rises, we should expect the equilibrium price of leather to ____ and the equilibrium quantity of leather to ____.
- A. rise; fall
 B. fall; rise
 C. rise; rise
 D. fall; fall
 E. fall; either rise or fall

Solution: B. When the demand for beef rises, the quantity of beef supplied rises as we move up the beef supply curve. As ranchers produce more beef they also produce more leather at any given price of leather. Leather supply moves out, resulting in a price decrease and a quantity increase.

- 15) [3 points] Clary, Jace and Izzy have 10 hours each to split between producing potions and steles. Clary can produce more steles in one hour than Jace and Izzy. Right now, Clary and Izzy are only producing steles, while Jace produces both potions and steles. Assuming efficient production, ____ has the comparative advantage in potions and ____ has the comparative advantage in steles.
- A. Jace; either Clary or Izzy
 B. Jace; Clary
 C. Jace; both Clary and Izzy
 D. Clary; Jace

E. either Clary or Izzy; Jace

Solution: A. As Jace is the first person put into potion production, he has the comparative advantage in potions. We don't have information on Clary's and Izzy's Opportunity Costs, so we don't know which of them has the comparative advantage in steles — we are only told that Clary has the *absolute* advantage in stele production, which is a different thing.

- 16) [3 points] When the *Washington Post* newspaper offered buyers a discount of 60% off the current subscription price of \$2.50, the number of subscriptions sold tripled. Demand is ____ and the lower price ____ revenues.

A. elastic; increased
B. inelastic; increased
C. unit elastic; did not change
D. elastic; decreased
E. inelastic; decreased

Solution: A. When price falls, elastic demand results in a more than proportional increase in quantity demanded. Here price fell 60% while quantity rose 200%, so demand is elastic, and revenue rises even though price falls: the new price is \$1 on three times the subscriptions.

- 17) [3 points] The perfectly competitive market for chalk has the typical downward-sloping demand and upward-sloping supply. Chalk is produced using calcium carbonate and is used in classrooms on chalkboards. When the price of calcium carbonate decreases, we observe that spending on chalk decreases. This outcome requires that market ____ for chalk is ____.

A. demand; elastic
B. demand; inelastic
C. demand; unit elastic
D. supply; either elastic, inelastic or unit elastic
E. supply; inelastic

Solution: B. When the price of an input decreases, the MC of production falls and supply moves out. We move down along the demand curve to a lower price and a higher quantity. If the impact of the lower price dominates, so that spending falls, demand must be inelastic.

- 18) [3 points] The market for sandwiches on the UofB campus is perfectly competitive. Consider the summer 2025 semester. Over summer, there are fewer students on campus, all else stays the same. Lucas is taking summer classes and has a downward-sloping MWTP for sandwiches. Compared to the fall semester, market price for sandwiches will ____ and Lucas buys ____ sandwiches over summer.

A. decrease; more
B. decrease; fewer
C. decrease; the same number of
D. increase; fewer
E. increase; the same number of

Solution: A. A decrease in market demand causes the market price to fall. Lucas' own MWTP is unchanged — he is one of the students still on campus. With the lower price and the same MWTP, more units pass his MWTP-versus-price threshold, so he buys more sandwiches.

- 19) [3 points] The market for peaches is perfectly competitive with downward-sloping demand and upward-sloping supply. Consumers choose between peaches and bananas. Based on this information, when the price of bananas increases, for the peach market Consumer Surplus ____ and Producer Surplus ____.

- A. increases; increases
- B. increases; decreases
- C. could increase or decrease; increases
- D. could increase or decrease; decreases
- E. decreases; decreases

Solution: C. An increase in the price of a substitute increases MWTP for peaches and shifts demand out, so market price and quantity both increase. As both MWTP and P increase, we don't know the net effect on Consumer Surplus. However, P increases while MC stays the same, so Producer Surplus increases.

- 20) [3 points] We have a perfectly competitive market with the typical downward-sloping demand and upward-sloping supply. As a result of changes in the market, we observe that market price stays the same, but market quantity decreases. As a result, we would expect consumer surplus to ____ and producer surplus to ____.
- A. increase; increase.
 - B. increase; either increase or decrease
 - C. decrease; either increase or decrease
 - D. decrease; decrease
 - E. stay the same; stay the same.

Solution: D. The only way to get a decrease in quantity with market price staying the same is a decrease in MWTP and an increase in MC whose price effects exactly offset each other. Since the price is unchanged, $CS = MWTP - P$ decreases — MWTP has fallen and fewer units trade — and $PS = P - MC$ decreases too, since MC has risen and fewer units trade.

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No work here will be considered.

Short Answer Questions

- Q1) Assume perfectly competitive markets with typical curves. At Halloween, consumers choose between two types of candy to give trick-or-treaters: chocolate bars and gummy candies. Chocolate bars are made with cocoa. Gummy candies do not contain any cocoa. A severe drought in cocoa producing areas has increased the price of cocoa.

- Q1a) [5 points] **Claim:** As gummy candy production doesn't use cocoa, the market price of gummy candy will not change when cocoa prices rise. Agree, disagree, or it depends? Explain. A diagram is not an explanation.

Solution: Disagree. The gummy market is linked to the chocolate market through demand, not through cocoa.

An increase in the MC of chocolate shifts chocolate supply up and increases chocolate prices. Consumers consider gummy candy and chocolate candy substitutes, since they choose between them. As the price of chocolate bars rises, people's MWTP — and hence demand — for gummy candies increases. As the demand curve moves out, we move up along the supply curve for gummy candy to a higher price for gummy candy this Halloween.

- Q1b) [6 points] Analysts predict consumer spending on both chocolate bars and gummy candy will increase when cocoa prices rise. What assumptions are they making about elasticities, if any, to reach this conclusion? Explain your reasoning. A diagram is not an explanation.

Solution: One assumption, about chocolate only. The gummy prediction needs no elasticity assumption at all.

Chocolate. We move up along the demand curve for chocolate when MC increases, so we need the elasticity of demand to predict the net impact on spending. If demand for chocolate is inelastic, the higher price dominates the lower quantity and spending rises. That is the assumption the analysts are making.

Gummy candy. Here the demand curve itself shifts out, so both price and quantity increase. Spending is $P \times Q$ and both go up, so no elasticity assumption is required to predict that spending increases.

Q2) People choose between streaming music online or buying vinyl records and listening to music on their record player. Harmonix is an online music streaming service. For \$11 a month, subscribers to Harmonix can listen to as many songs as they want, with no additional payment. Consumers are different, but every consumer's marginal willingness to pay for songs streamed is decreasing in the number of songs every month. Harmonix pays artists \$0.01 every time their song is streamed.

Q2a) [5 points] Vinyl records have suddenly become more popular thanks to social media. Assume subscribers continue to buy the \$11 a month Harmonix subscription. Does the increased popularity of vinyl affect the number of songs they stream a month? Why or why not? Explain your reasoning.

Solution: Yes — they stream fewer songs, even though nothing about the price has changed.

The price of an extra streamed song is zero, and it remains zero. But if the next best option to a streamed song — listening to it on vinyl — becomes better, the MWTP for streamed songs decreases at every quantity. At the same price of zero for an extra song, when MWTP decreases fewer units cross the MWTP-versus-price threshold. Consumers will choose to stream fewer songs a month than before.

Q2b) [6 points] In December, Harmonix plans to raise the subscription price to \$14 a month. **Claim:** Artists will earn less money from Harmonix because of fewer streams. Agree, disagree, or it depends? Explain your reasoning.

Solution: Agree — but the streams are lost because people leave, not because the people who stay listen less.

Each individual who chooses to continue subscribing faces the same price of an extra song, zero, and their MWTP is unchanged, so they choose exactly the same number of songs. Same quantity means their total benefit is the same, while their total cost is now higher, since the fixed subscription price is higher.

So there will now be people for whom, at the same quantity as before, total benefit is no longer enough to cover the higher \$14 subscription fee. They will not subscribe to Harmonix any more. It is those lost subscribers, not reduced listening by the ones who stay, that reduce the number of streams and hence artists' earnings.

Note: Some of you used the scenario from (a) here as well. We did not penalize you for that. However, we were looking for an explanation of how the \$14 price specifically affected the quantity of streams in your answer.

Q3) Assume no externalities. Beyoncé is doing a New Year's Eve show at the Scotiabank Arena, which has a strict capacity of 20,000 seats. The posted price is \$500, and expecting excess demand, all of those who want to purchase at \$500 enter a lottery. 50,000 entered the lottery. 20,000 of them were randomly chosen: they received a ticket and their credit card was charged.

Q3a) [5 points] **Claim:** A higher ticket price would have increased total surplus. Agree, disagree, or it depends? Explain your reasoning.

Solution: It depends — on how much higher.

Without externalities, if we move a ticket from a lower-MWTP buyer to a higher-MWTP one who didn't get a ticket, total surplus increases. Increasing the price means buyers with a lower MWTP will no longer want to buy, leaving only higher-MWTP buyers in the pool. If 20,000 are then chosen from this

smaller, higher-MWTP pool, it is *likely* — though not guaranteed, since it is still a lottery — that we end up with a higher surplus from the same 20,000 tickets.

If you raise the price so that exactly 20,000 people want to buy, only the highest-MWTP people get tickets and they are allocated efficiently. Raising the price beyond that point would leave seats empty and decrease Total Surplus.

- Q3b)** [5 points] Xin, who was willing to pay \$2,000 for the ticket, did not get a ticket from the lottery. Yana, who was willing to pay \$750 for the ticket, did get a ticket from the lottery. **Claim:** Allowing Yana to sell her ticket to Xin would increase surplus by \$1,500. Agree, disagree, or it depends? Explain your reasoning.

Solution: Disagree. The trade does increase surplus, but by \$1,250, not \$1,500.

If Yana doesn't sell the ticket to Xin, her next best option is attending the concert and getting \$750 in value. If she sells it to Xin she needs to get at minimum \$750 from him, and Xin is willing to pay up to \$2,000. There is room for them to negotiate a price between \$750 and \$2,000 to move the ticket from Yana to Xin.

When this happens, society loses the \$750 Yana would have received had she gone to the concert, but gains the \$2,000 Xin gets from going instead. Total surplus increases by $\$2,000 - \$750 = \$1,250$. We cannot determine how the *individual* surpluses change, because we don't know what price the trade happens at — but that price is a transfer and doesn't affect the total.

- Q3c)** [5 points] Assume that any ticket resale must take place at the posted price of \$500 and this policy can be enforced. Explain how, if at all, this policy affects your response to part (b).

Solution: It kills the trade entirely, so total surplus is unchanged.

From part (b) we know that for the trade to happen the price needs to be between \$750 and \$2,000. With the \$500 resale price restriction, Yana won't sell the ticket: she is better off just attending the concert and getting \$750 of value. As the trade will not happen, total surplus remains unchanged — the \$1,250 gain identified in part (b) is simply not realized.

- Q4)** Upland and Downland produce chickpeas and oranges with their land and trade with each other. Upland exports chickpeas to Downland and imports oranges. The exchange rate is 1 kilo of chickpeas for 1 kilo of oranges, which makes both countries better off.

- Q4a)** [5 points] Two observations: (1) With the same land, Upland produces more oranges than Downland. (2) Upland exports chickpeas to Downland. How can we understand these two observations?

Solution: There is no contradiction: trade patterns are determined by comparative advantage, not absolute advantage.

Observation (1) just tells us that Upland has the absolute advantage in oranges. Observation (2) tells us Upland has the comparative advantage in chickpeas — that is, a lower Opportunity Cost of producing them. Upland's productivity at chickpeas must therefore be such that

$$OC_{chickpeas}^{Upland} = \frac{\text{oranges per hour}}{\text{chickpeas per hour}}$$

is lower than Downland's. A country can be better at producing everything and still import the good it is comparatively worse at.

- Q4b)** [6 points] Trade with Downland is no longer possible. Upland must now produce any chickpeas and oranges it wants by itself. **Claim:** The cost of 1 kilo of oranges in Upland increases when trade stops. Agree, disagree, or it depends? Explain your reasoning.

Solution: Agree.

Without trade, Upland has to produce its own oranges at a cost of

$$OC_{oranges}^{\text{Upland}} = \frac{\text{chickpeas per hour}}{\text{oranges per hour}}$$

With trade, the reason Upland was buying oranges is that it was getting them at a price cheaper than it could manage at home — that is why it benefited from trade. It must therefore be that

$$OC_{oranges}^{\text{Upland}} > 1 > OC_{oranges}^{\text{Downland}}$$

Thus when trade is no longer possible, Upland's Opportunity Cost of 1 kilo of oranges increases from 1 (the exchange rate it used to get) to $OC_{oranges}^{\text{Upland}}$, which is higher.